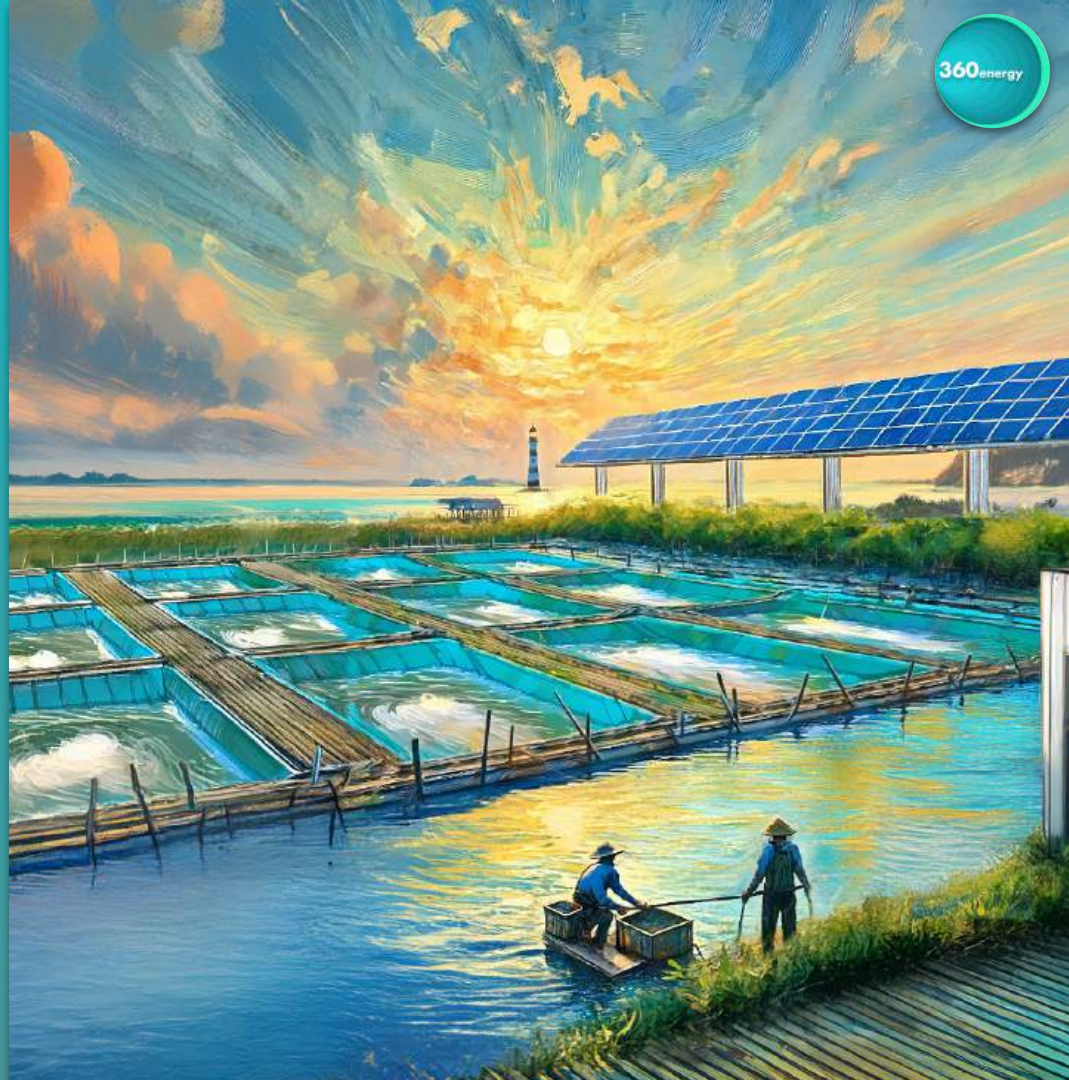


360energy

Company Overview

March 2025

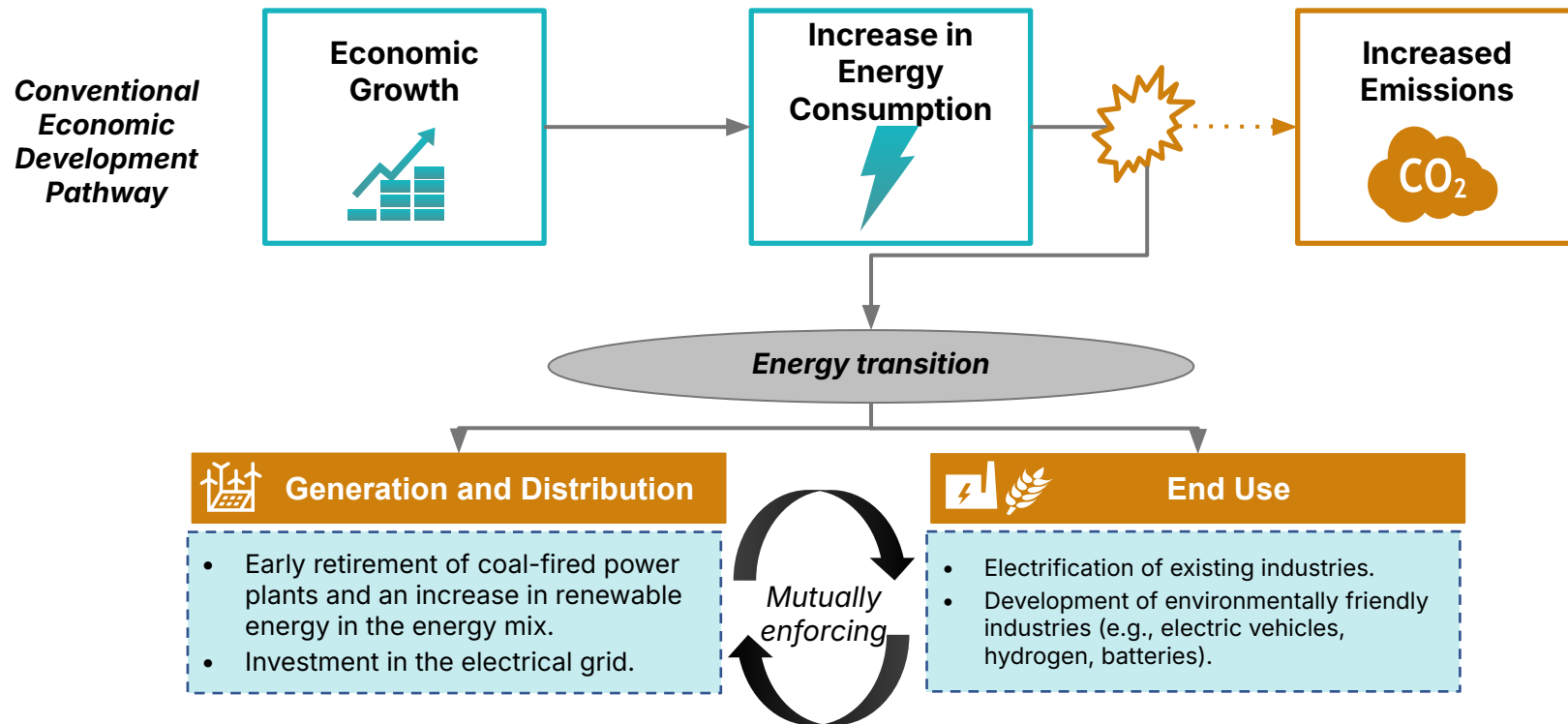


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Background

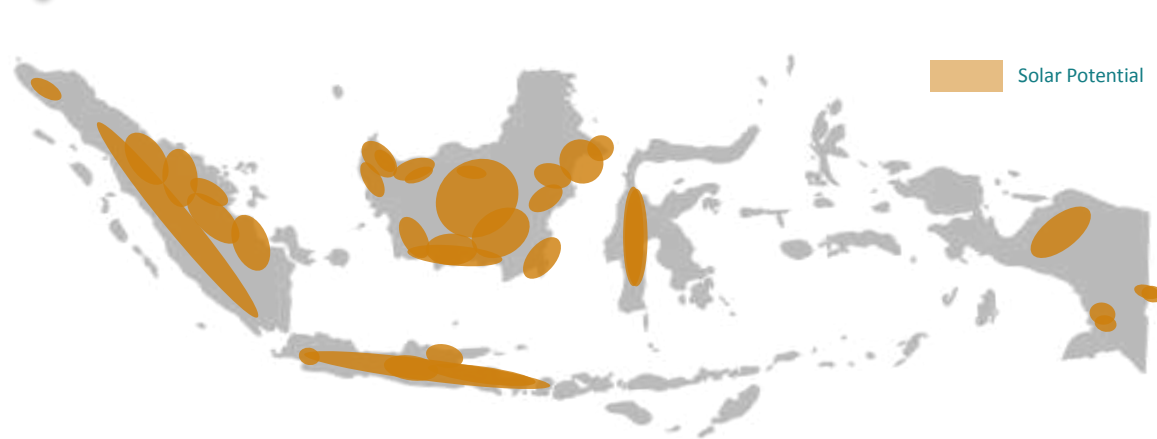
Indonesia holds the potential for **rapid industrialization powered by clean energy**

Pathway to Sustainable Industrialization



Background

Indonesia's **massive solar energy potential and strong regulatory framework** provides a strong foundation for widespread solar adoption



Solar Potential
Nationally

Has been
utilized today

Cost of solar panels in
Southeast Asia have
dropped by **89%** in the
past 10 years (Get Solar)



Power Purchase Agreement

Permen ESDM No.4/2020

Privately produces power for PLN grid



Commercial Solar BOT

Permen ESDM No.2/2024, No.50/2017

Solar installation for on-site consumption



Green Distributed Energy

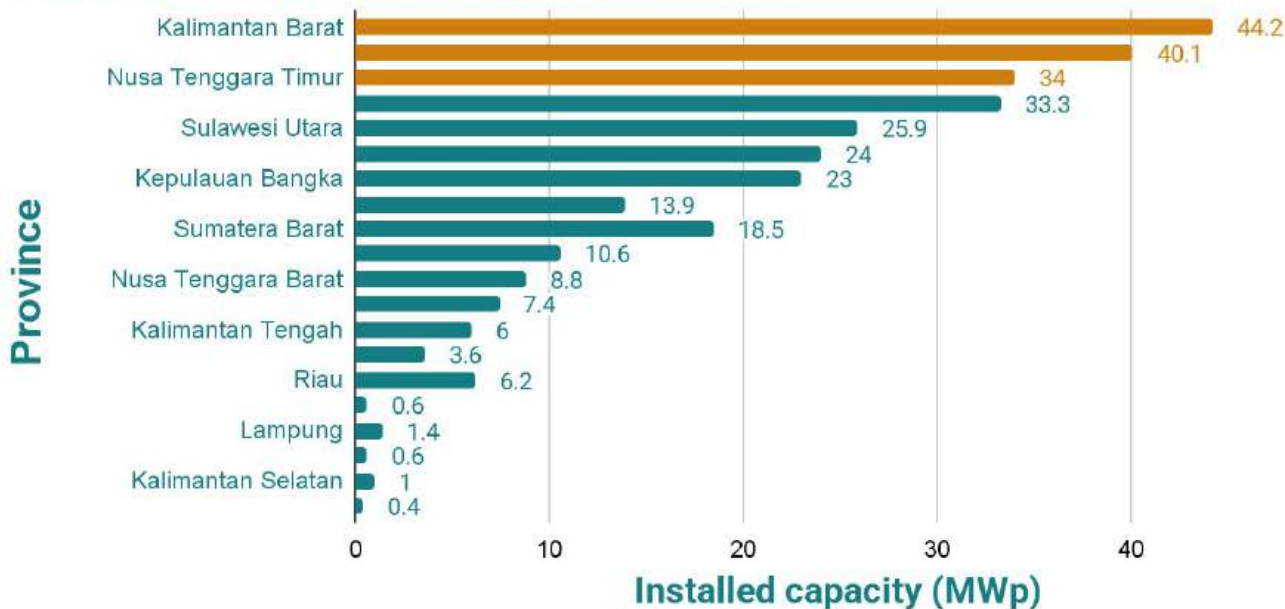
RUPTL 2021-2030, Permen ESDM No.4/2020

Facilitates rural dieselisation in mini-grids

Dedieselization Overview

Distributed demand for de-dieselization can **reduce fuel costs, grow rural industries and support energy independence** (swasembada energi)

Projected Dedieselisasi Demand Per RUTPL



Decarbonizing rural electricity grids can **support energy independence** (swasembada energi)



Solar Power can **power sustainable industries** in rural areas that are export ready



Emissions reductions from solar can **generate carbon credits** (if not at cost to diesel)

National Dedieselization has a Serviceable Addressable Market (SAM) size of **US\$ 235.4 Million**

Company Overview

Our approach enables companies to **decarbonize operations with Zero CapEx** achieved through Joint ventures and research partnerships



Established Expertise: Strong foundation with Solar BOT (Build-Operate-Transfer) as an entry point into Indonesia's renewable energy market.



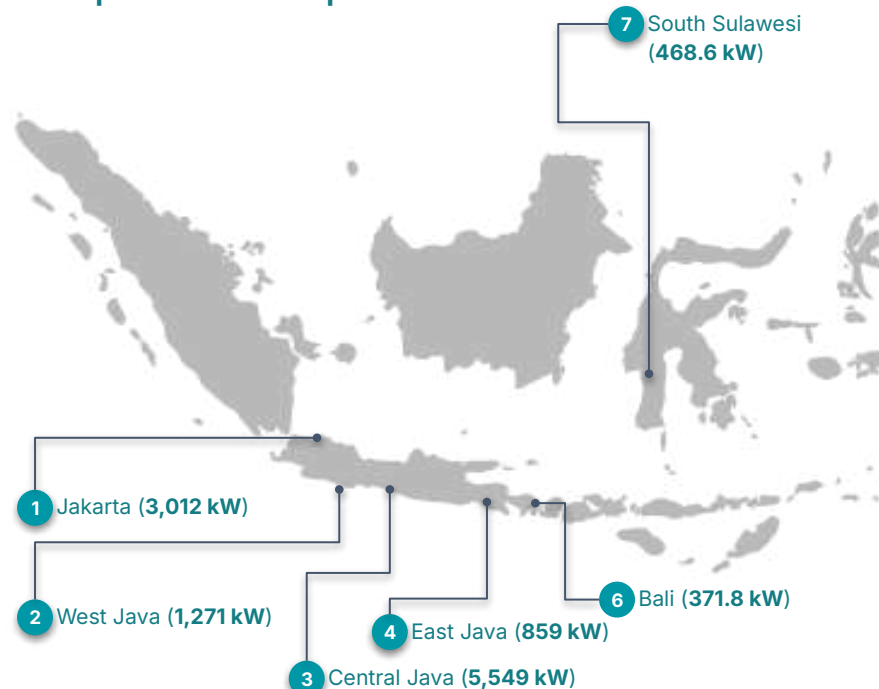
Commitment to Decarbonization: Focused on scaling renewable energy adoption through SME-friendly financing models.



Long-Term Partnerships & Research: Established strong research ties with BRIN, POLBAN, and international institutions like WPI, MIT to integrate solar, aquaculture, and battery technologies into scalable models



Growth Potential & Market Impact: Positioned at the intersection of sustainability, energy transition, and SME empowerment



Total Installed Capacity 11,650 kW

Growth Strategy

We drive industrial decarbonization by delivering **cost-effective solar BOT projects and pioneering solar-enabled solutions for high-growth SMEs**

Commercial Solar BOT

Reduces electricity payments through solar leasing



10% Cheaper Electricity



Zero CapEx

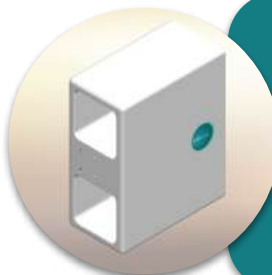


Free Operating & Maintenance

Solar-enabled Projects



Solar BioFloc :
Solar-powered super intensive farming with BOT model



Solaris BSS:
Universal Battery Swapping Station for Motorcycle EVs

Solar BOT



BOT: sample project

Solar energy can cover **up to 17% of total facility energy** needs without the need for battery storage and reduce upto 10% in energy spending



Step 1: Site Assessment & Detailed Engineering Design (DED)



Step 2: Financing & Permitting



Step 3: Installation & Operation



Step 4: Flexible End-of-Contract Options

BOT: sample project

Solar energy can cover **up to 17% of total facility energy** needs without the need for battery storage and reduce upto 10% in energy spending

Minimum Project Size:



Jabodetabek
<100 kWp

PSH (Peak Sun Hours):

The number of hours per day with peak solar intensity is typically 4-5 hours in most locations



Java
<100 kWp

Direct Consumption:

Solar power is used immediately during peak hours, reducing grid dependency



Outside Java
<100 kWp

Cost Savings:

Reduces electricity costs by 10-15%, providing predictable energy pricing.

BOT: Overview

The BOT model offers a 25-year turnkey solution with zero upfront costs, energy savings, and flexible options post-contract



Contract Period: 25 Years

A long-term commitment ensuring stable and predictable energy costs for the client over two and a half decades.



10% Reduction in Energy Tariff

Clients benefit from a guaranteed 10% reduction compared to PLN electricity tariffs, making renewable energy financially attractive.



Zero CapEx

No upfront capital expenditure required, all initial costs, removing financial barriers to adopting renewable energy



Minimum Capacity: >100 kWp

The BOT model is available for installations of >100 kWp for Jawa-Bali-Madura, >250 kWp for outside of JAMALI grid ensuring economies of scale and sufficient impact on energy costs



Free Operation & Maintenance (O&M):

Project developer handles all maintenance, repairs, and monitoring for the full 25 years, ensuring peak performance without added responsibilities for the client

BOT: sample project

We have an extensive suite of projects developed through partnerships with dedicated OEMs and EPCs

PDAM Makassar



468.6 Kwp
Installed Capacity



469.75 Tonnes
Co2 avoided



1,052 trees
Equivalent Planted

PDAM Makassar



BOT: sample project

We have an extensive suite of projects developed through partnerships with dedicated OEMs and EPCs

Widodo Makmur Perkasa



5,147.04 Kwp
Installed Capacity



349.5 Tonnes
Co2 avoided

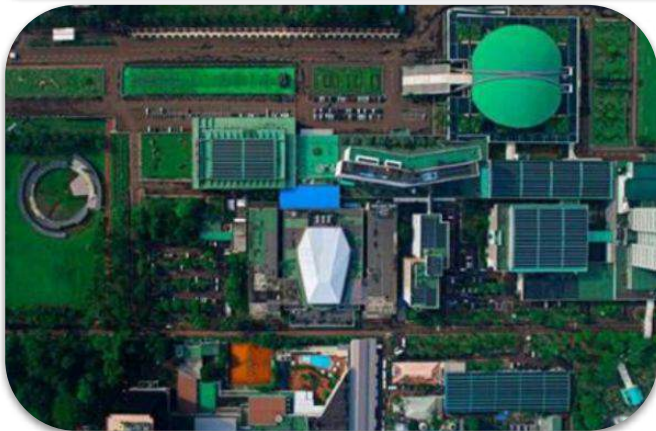


10,985 trees
Equivalent Planted

BOT: sample project

We have an extensive suite of projects developed through partnerships with dedicated OEMs and EPCs

DPR RI



1590 Kwp
Installed Capacity



656.6 Tonnes
Co2 avoided



1476 trees
Equivalent Planted

BOT: sample project

We have an extensive suite of projects developed through partnerships with dedicated OEMs and EPCs

Angkasa Pura Group



1276.6 Kwp
Installed Capacity



866.8 Tonnes
Co2 avoided



36,116 trees
Equivalent Planted

BOT: sample project

We have an extensive suite of projects developed through partnerships with dedicated OEMs and EPCs

Pakuwon Group



PAKUWON GROUP



1718.3 Kwp
Installed Capacity



1,166 Tonnes
Co2 avoided



48,614 trees
Equivalent Planted

Solar BioFloc



BOT: sample project

Traditional aquaculture systems face challenges in productivity, sustainability, and market readiness, leading to inconsistent yields and lower returns

Description	Rice Field (Minapadi)	Traditional Intensive Pond	Flow-Through System (KAD)	Biofloc System	IBK RAS
System Type	Extensive	Intensive	Intensive	Intensive	Super-intensive
Stocking Density	0.25 kg/m ²	2.5 kg/m²	18.75-25 kg/m ³	25 kg/m³	150 kg/m³
Stock Size	7-8 cm	5-8 cm	8-12 cm	8-10 cm	8-10 cm
Culture Duration	3 months	3-4 months	3 months	3 months	3 months
Harvest Size	150 grams	250 grams	300 grams	200-300 grams	200-300 grams
FCR (Feed Conversion Ratio)	1.0	1.5	1.7	1.0	1.0
Productivity	0.15 kg/m ²	2 kg/m²	30 kg/m ³	20-30 kg/m³	TBD
Cost per kg (IDR)	9,500	14,250	16,150	10,500	TBD
Gross Profit (IDR/m ²)	1,275/m ²	7,500/m²	55,500/m ³	190,000 - 285,000/m³	TBD

Comparison of Red Tilapia Cultivation Methods (BBPAT Sukabumi) (White, 2008)

Sukabumi farm currently operates a traditional intensive pond system (red)

Low Productivity: Poor growth rates and inefficient feed conversion (high FCR)

Unsustainable: High energy use and poor water quality management increase costs.

Market Limitations: Inconsistent product quality limits access to higher-value export markets.

Many of these problems can be solved by utilizing a super-intensive biofloc system (green)

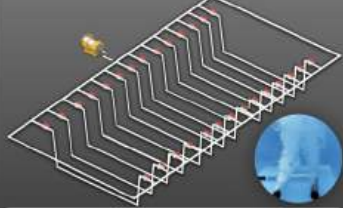
Solar Biofloc: sample project

Solar Bubbler offers operational efficiency, higher quality export-ready products, and stable market access, distinguishing us from traditional methods

Solar Bubbler

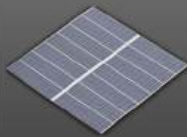


Root Blower + Diffuser



Delivers airflow to maintain optimal oxygen levels for fish growth. Evenly distribute oxygen across the pond for uniform water quality

Solar PV



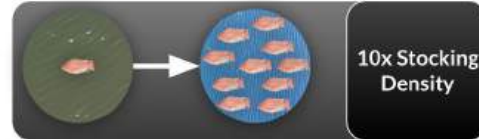
Power the system with renewable energy, cutting costs and emissions, empowering rural farms

Internet of Things (IoT) Systems

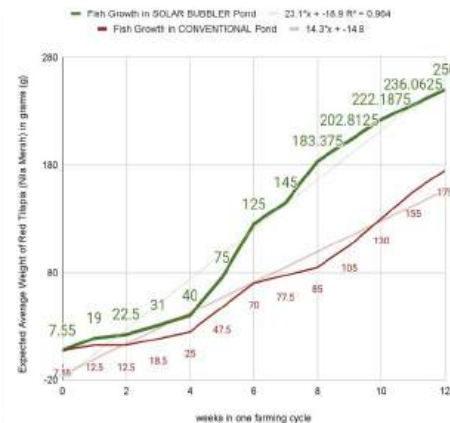


Maintain ideal growing conditions through sensors and auto feeder

Sustainable Farming SOP



Red Tilapia Growth Increase with Solar Bubbler System



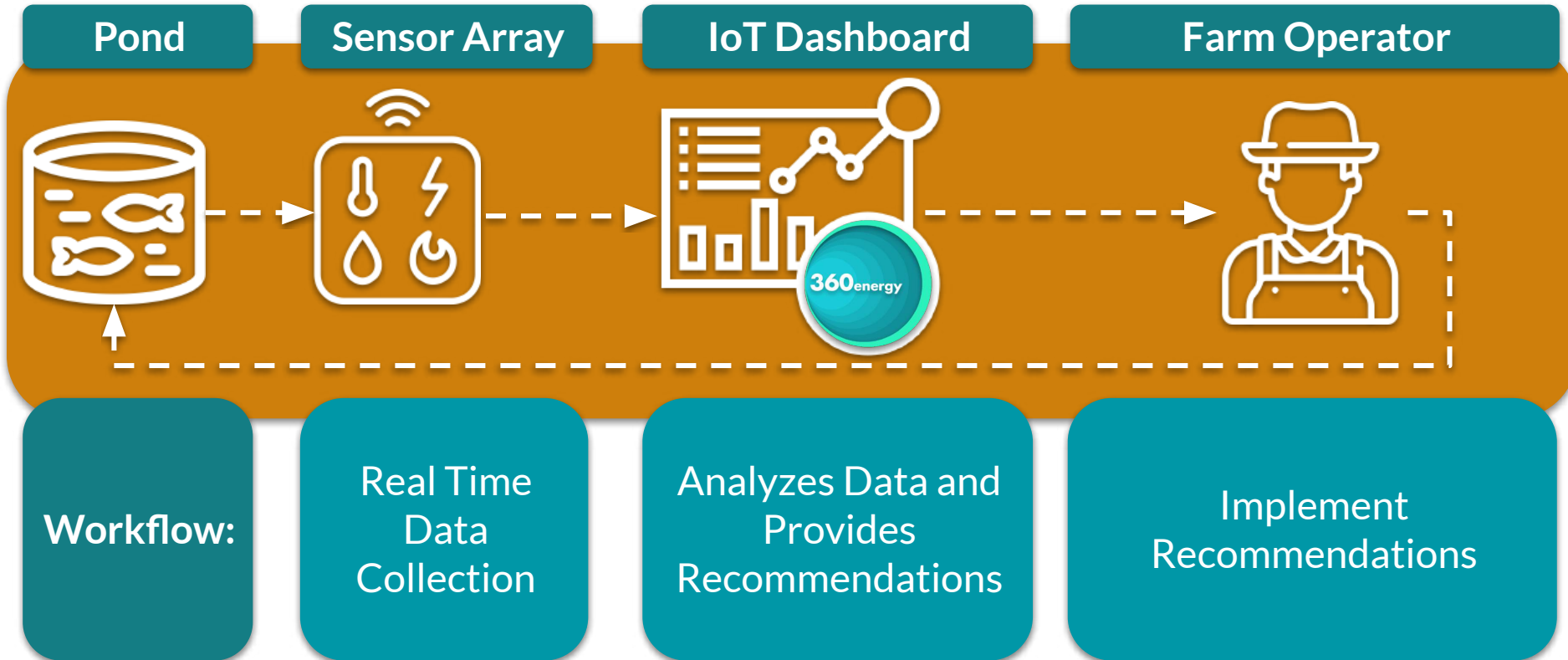
61.82% Increase in Individual Fish Growth



Export Ready Tilapia and Vannamei Shrimp

IoT Workflow

This continuous feedback loop, facilitated by the IoT dashboard, helps farmers maintain optimal conditions, improve productivity, and ensure the sustainability



Solar Biofloc: Project Implementation

Solar bubbler has worked to increase stocking density (padat tebar) to 150kg/M3 while reducing electricity spending from solar

Sukabumi Project Site



70 Kwp
Installed Capacity



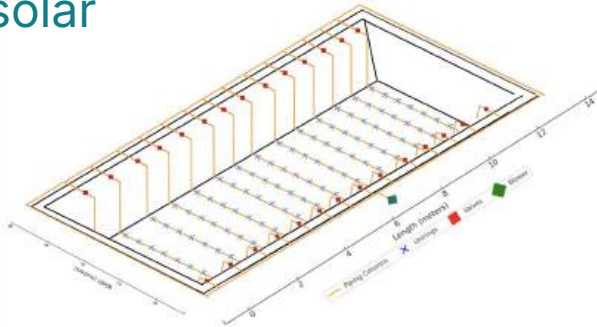
4.4 Tonnes
Co2 avoided



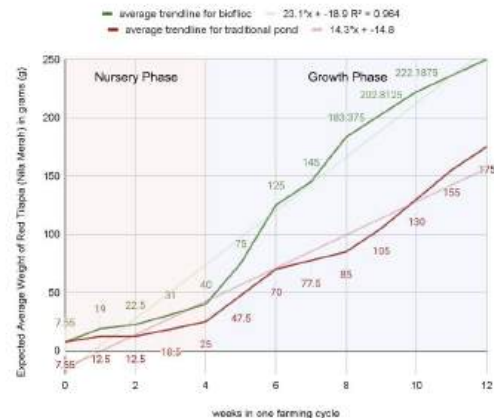
1-1.2 FCR
With Biofloc



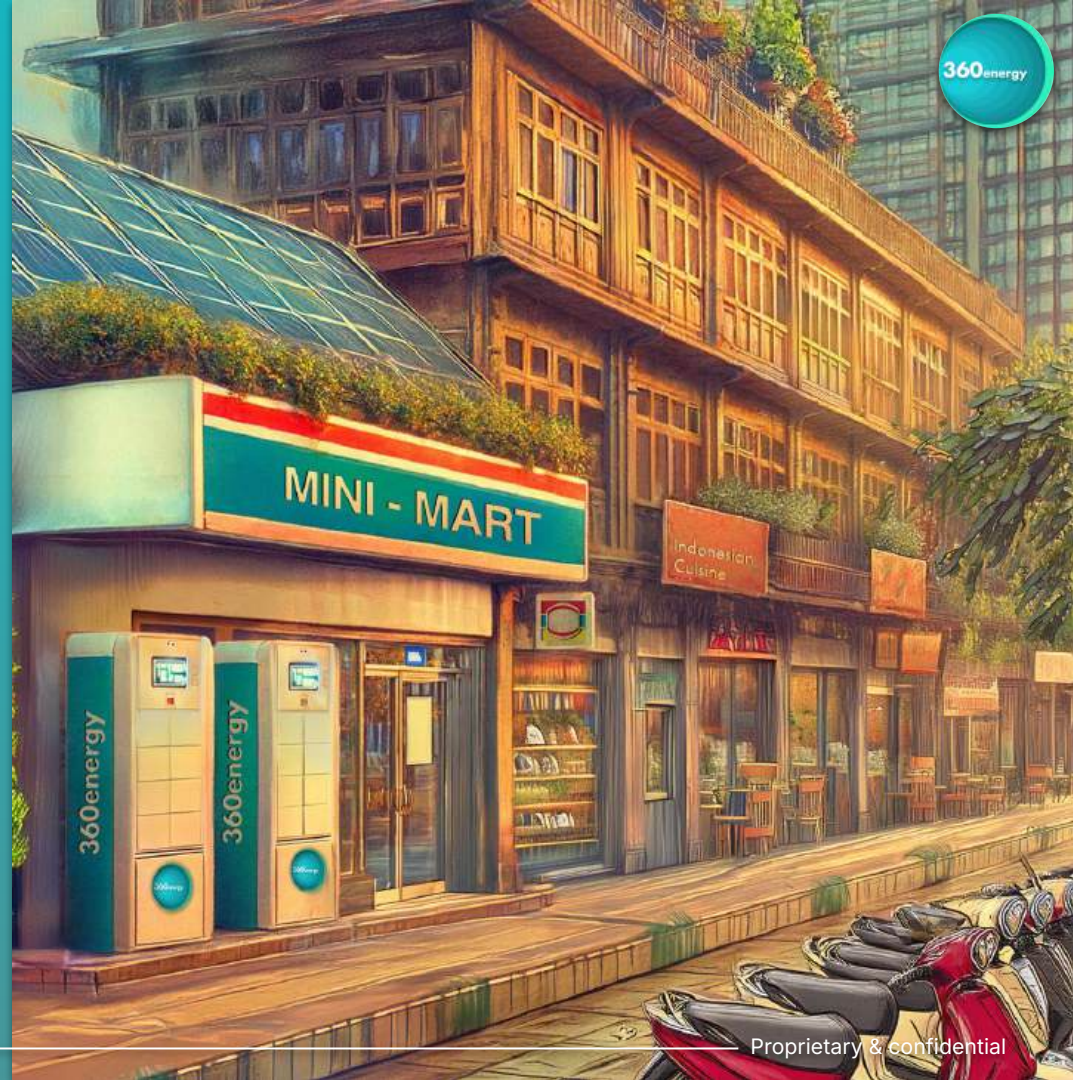
150 kg / M3
Stocking Density



Graph A: Comparative Analysis of Biofloc V. Traditional intensive pond farming growth targets



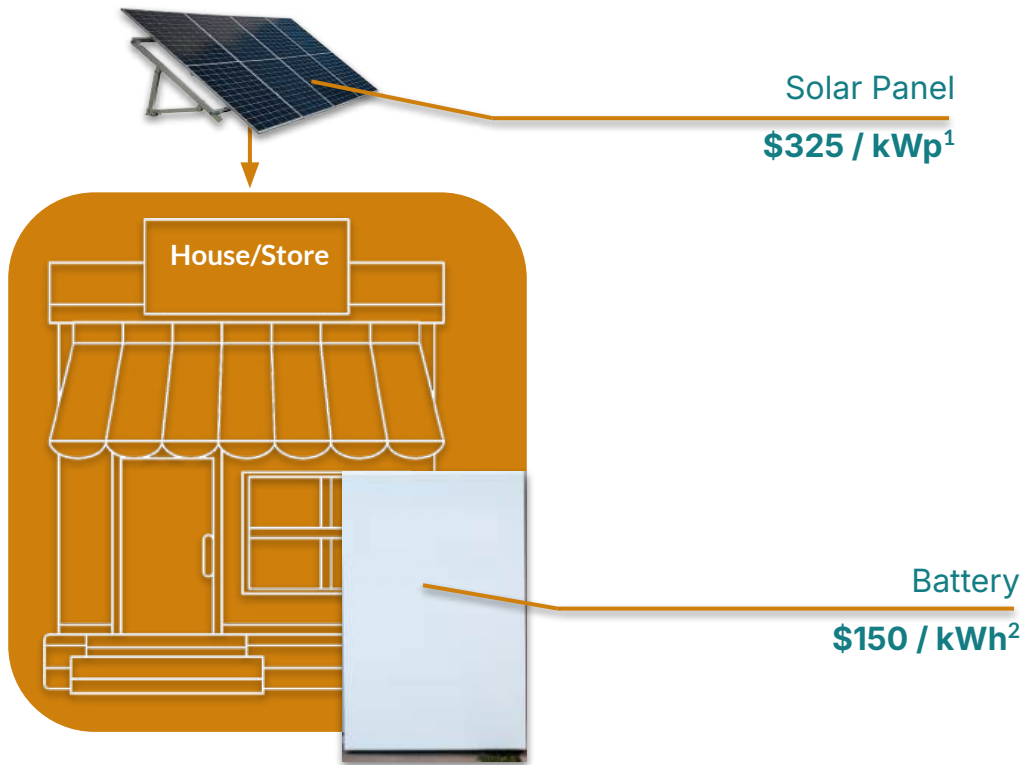
Solaris BSS



360energy

The Problem

Indonesia holds massive potential for solar power, yet high costs of battery storage hinders distributed solar adoption



The need for battery increases the cost of installation by 287.5%²

There is a need to decrease cost of battery storage for Distributed solar

OUR SOLUTION: *Universal* Battery Swapping Station (BSS) for Solar Energy Storage

Solaris BSS

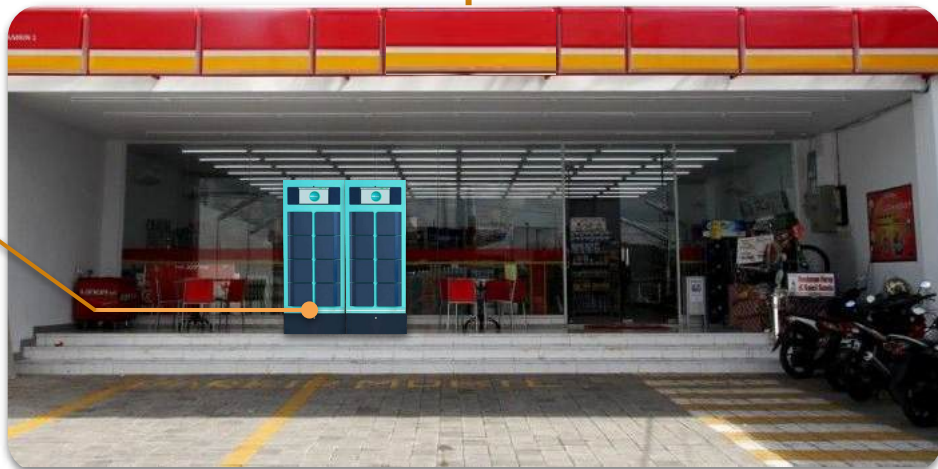
23



Solaris acts as an energy storage system for solar panels

Reduces the overall cost of installing solar panels

Helps expand EV charging infrastructure



Indonesian Convenience Store

Motorcycles with swappable batteries provide a solution for slow-charging, range anxiety, and battery depreciation associated with owning EVs



Typical EV motorcycles take **4 hours** for a full charge in public chargers



Swappable batteries take **30 seconds** to replace in public Battery Swapping Stations (BSS)

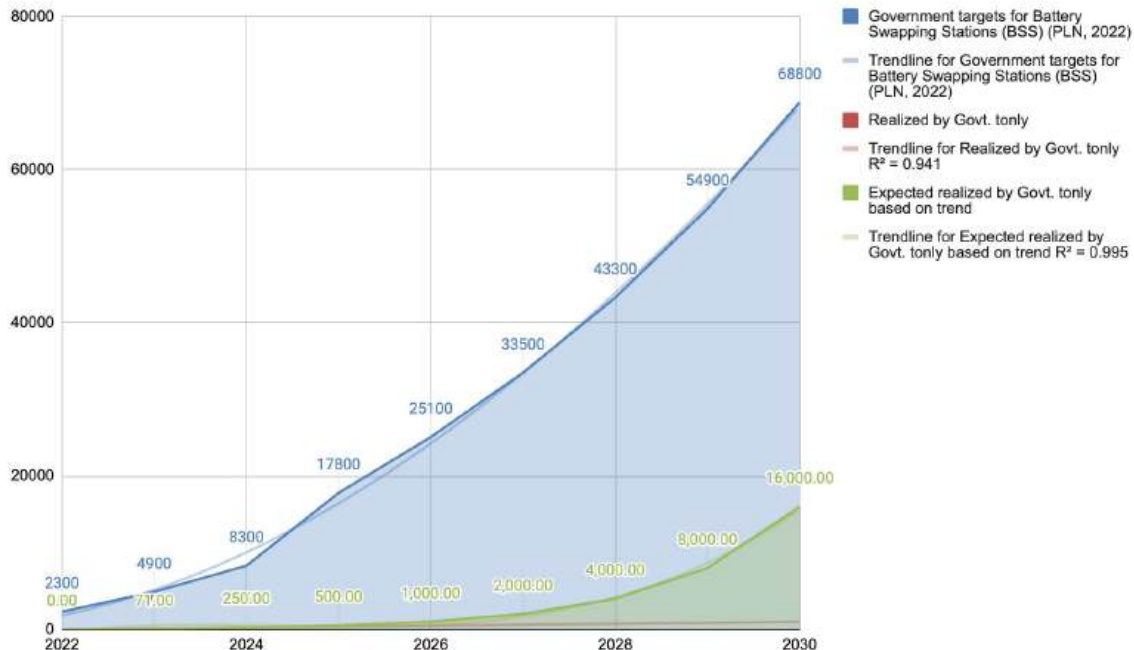


Taiwan and China has developed massive BSS networks, Southeast Asia is working to replicate this

BACKGROUND

Private intervention to develop EV infrastructure is needed to meet govt. targets of 9 million two-wheeler EVs by 2030s

GRAPH 1: Target Battery Swapping Stations Constructed Each Year



Two wheeler EV demands in SEA are projected to grow at **50% CAGR** in the coming years (East Ventures)

Govt. and SOEs can only fulfil partial charging demand (green)

Substantial Govt. support for domestic EV manufacturing lead to high demand for swaps (blue)

Analysis based on reports from State Electricity Company (PLN), Ministry of Energy and Mineral Resources (MEMR), Ministry of Maritime Affairs and Investment (MMAI) and impact reports from Swap.id, Oyika, and GesITS

BACKGROUND

Private intervention to develop EV infrastructure is needed to meet govt. targets of 9 million two-wheeler EVs by 2030s

Day use

(Optimized for battery swapping)

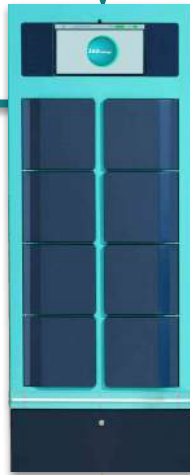


Demand for battery swapping peaks at commuting hours

Solar energy charging during day reduces emissions of battery swaps



All available solar energy is funneled to Solaris BSS



Subsidized Grid power charges BSS at all other times at 5 USc/kWh

Night use

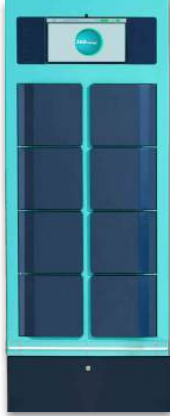
(Optimized for powering stores)



Power essential systems at night

Reduce overall utility bill for stores

Cabinet Leasing (CL) provides a **scalable and universal** opportunity to expand BSS network across different operators, with minimal expenses

	Motorcycle	Retrofitted Cabinets	Battery
Asset			
Stakeholder	Owned by driver or leased by Financier	Owned by 360energy Leased per-charge to Station operator	Operated by Station Operator

Expanding a BSS network is highly capital-intensive, with significant costs incurred from building and maintaining infrastructure

40%

of the cost of installing new stations tied to cabinet and logistics are **reduced through this model**

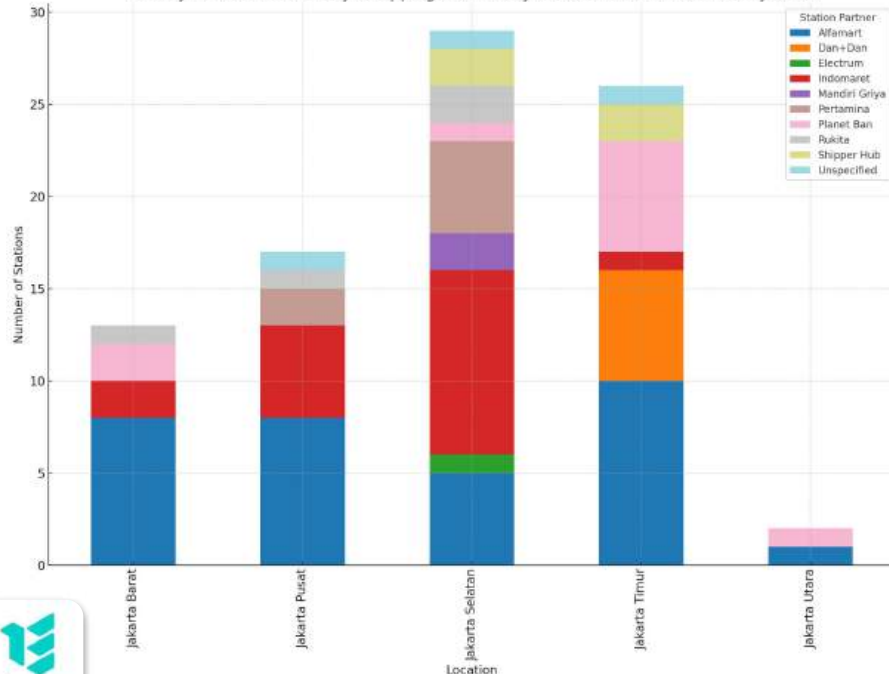
We make it easier to grow the BSS networks of station operators

We provides a scalable and universal model that can be **expanded across different networks** to further reduce costs and improve service delivery

BUSINESS MODEL: Cabinet Leasing (CL)

Solaris BSS **accelerates EV infrastructure growth** and facilitates CAPEX Reduction through strategic infrastructure integration

Density of Electrum Battery Swapping Stations by Partner Across Locations in Jakarta



Analysis of 87 Electrum Battery Swapping Stations throughout Jakarta

Station Operators collaborates with multiple partners to deploy BSS. This diversity in partnerships highlights the complexity and added operational burdens in scaling the network efficiently.

360energy proposes purchasing cabinets from station operators, retrofitting them with our scheduling algorithm, and installing them as energy storage systems to optimize CAPEX.

Reduced CAPEX: Demonstrating significant savings by leveraging existing infrastructure.

Scalability: Providing a model for cost-effective network expansion.

Sustainability: Supporting renewable energy integration and sustainability goals.

BUSINESS MODEL: Cabinet Leasing (CL)

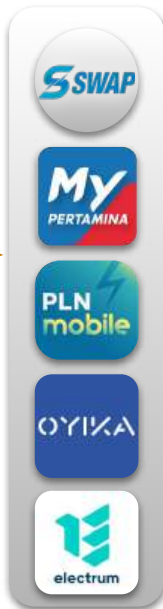
Solaris BSS **accelerates EV infrastructure growth** and facilitates CAPEX Reduction through strategic infrastructure integration

Cabinet Leasing



Riders pay Rp. 333/km

Rp12.6 mn /month in profits



\$

Pay-per-use



\$

Pay-per-month

Energy-as-a-service



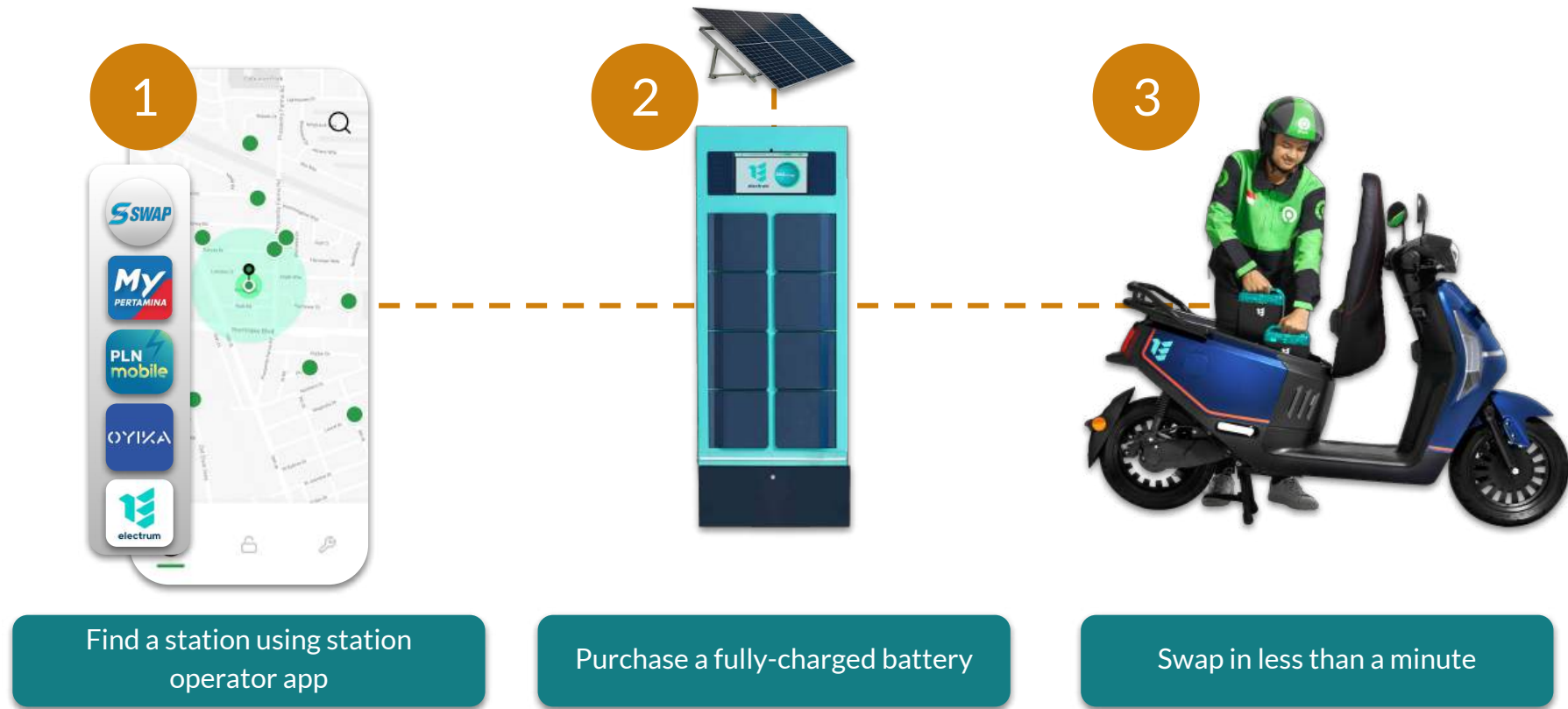
Solar energy leasing

8% reduction in monthly utilities

BUSINESS MODEL: Cabinet Leasing (CL)

Partnering with Station Operators

mean we provide a sleek and familiar experience for end-users

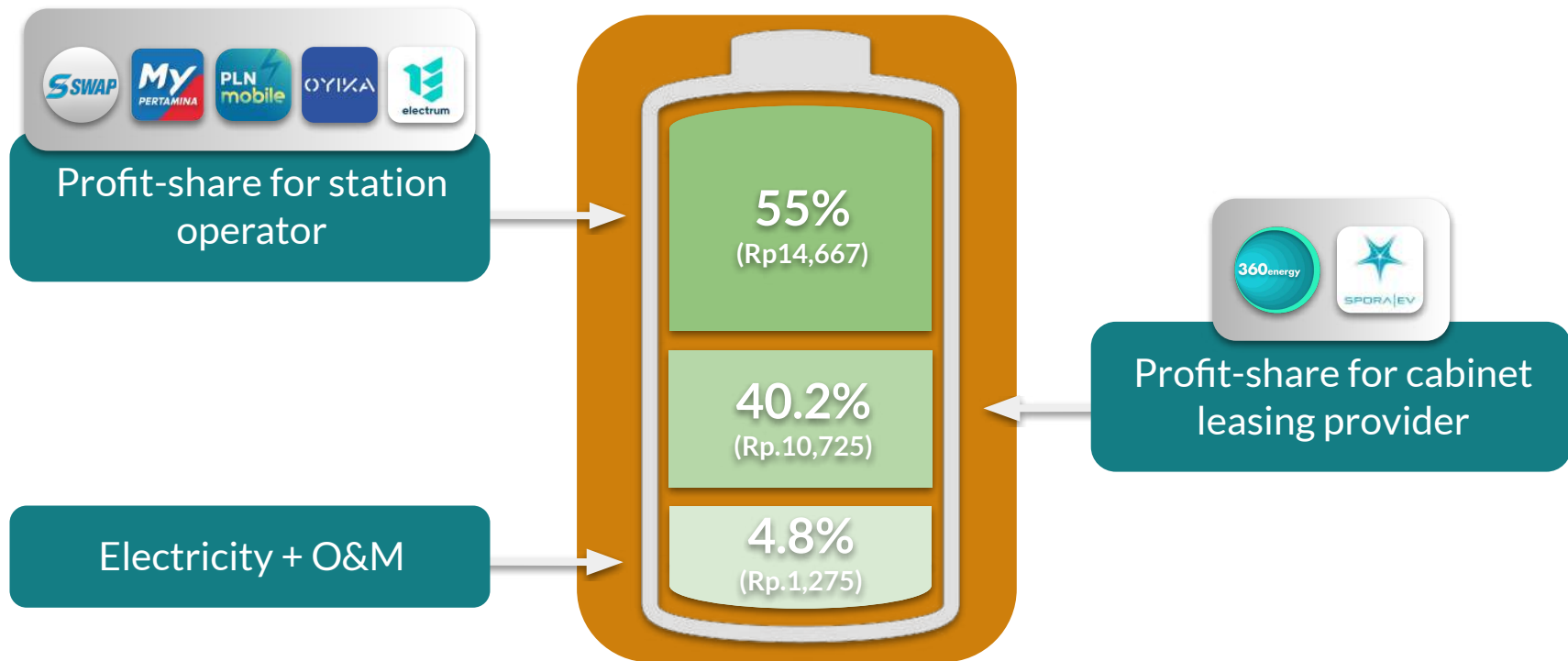


Our pricing strategy is affordable with a healthy Return on Investment (ROI)

IDR 26,667 price per swap

80km range

IDR 333/km



Consistent swapping revenues achieves **Break even point of 11 months**



IDR 26,667 price per swap

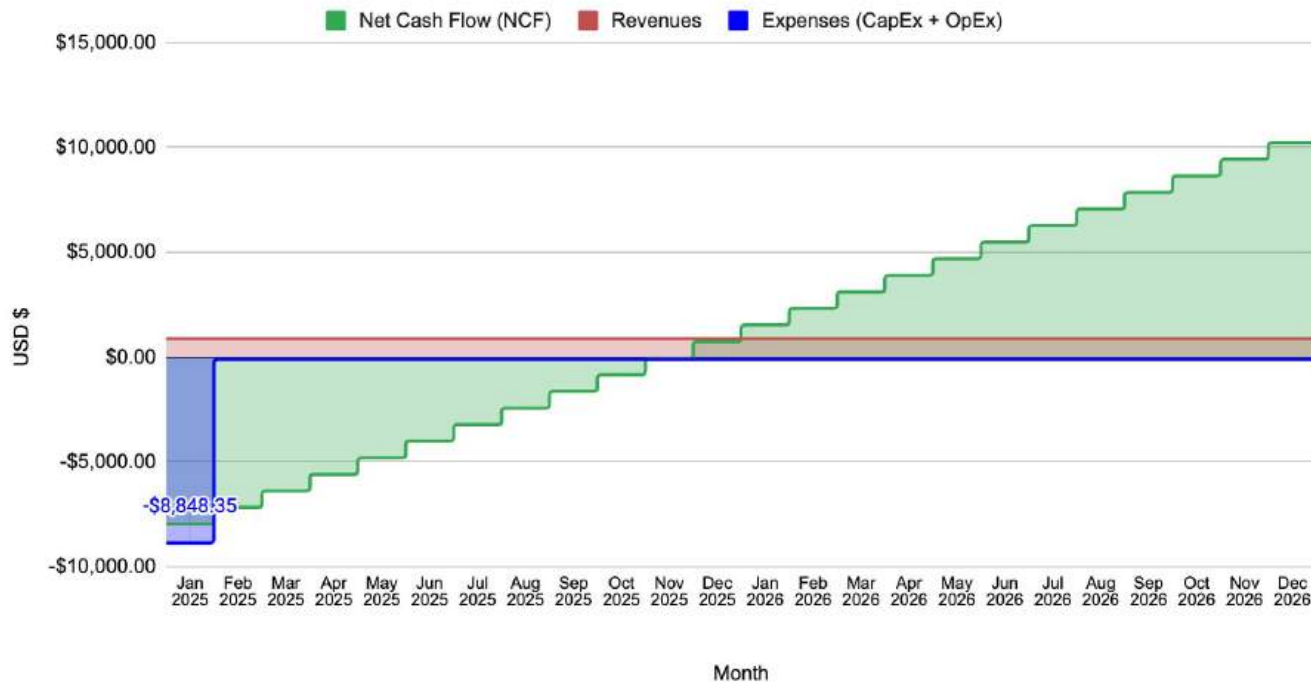
42 swaps per day

Monthly Profits of Rp12,6 mn

Break Even Point (BEP) 11 months

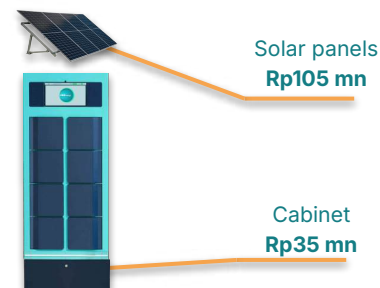
Each Battery Station provides a Healthy Return on Investment (ROI)

2YR Financial Analysis of a Single Solaris Battery Swapping Station (BSS)



Break Even Point of
11 months

Initial Capex:
Rp.140 mn



Annual Revenue:
Rp172 mn

Massive demand for battery swapping services

US\$
160
Million

Annual demand for battery swaps
2024-2030
(Total Addressable Market)

US\$
91.4
Million

Demand unfulfilled in 2026 by current
operators at current growth rate
(Serviceable Addressable Market)

US\$
2.9
Million

4% of ride hailing motorcycle demand
(Serviceable Obtainable Market)

Our first target market is EV Motorcycle drivers in Jakarta, Indonesia



Capital city of Indonesia with a metro population of **32.5 million¹**



Around **17.3 million²** motorcycles in use, **85% of all vehicles³**

Thank you

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Website: 360energy.io

Instagram: [instagram.com/360e.io/](https://www.instagram.com/360e.io/)

